

Lab 03: Diodes — Clipping and Clamping Circuits

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1 Abstract

This experiment studies wave-shaping using silicon diodes configured as clippers (series and parallel, with and without DC bias) and as clampers (DC restorer). You will measure output levels for positive and negative input excursions, verify predicted clip thresholds that include diode forward drop(s) and any DC offset sources, and evaluate clamper operation over frequency via the RC time constant criterion. Results will be summarized with scope captures and compared against hand analysis and optional SPICE simulations.

2 Introduction

Diode clipper networks bound a signal by conducting during one polarity to clamp the output to approximately a reference level (often a DC source plus diode drop) and otherwise allow the input to pass. Clamping circuits shift a waveform vertically without altering its peak-to-peak amplitude, provided the capacitor holds charge over a cycle (i.e., $RC \gg T$). We will implement representative topologies and confirm the expected transfer characteristics experimentally.

3 Objective

- Build and test biased/unbiased series and parallel diode clippers; record clip levels for both input polarities.
- Build a DC clamper and examine the effect of frequency on output shape; relate to the RC time constant.
- Compare measurements to predictions that account for V_D (silicon $\approx 0.7\text{ V}$) and bias sources; optionally validate with SPICE.

4 Theory

4.1 Clipping Circuits

A single diode with a series resistor forms a basic clipper. With a DC source V_X in series (or anti-series) with the diode, the output clips near $\pm(V_X + V_D)$ depending on orientation. For an ideal diode the transition is abrupt; a real diode presents $V_D \approx 0.6\text{ to }0.8\text{ V}$ and finite slope.

For biased parallel clippers (diode + source to ground), the output follows input until the diode forward-bias condition is met; then the output is held roughly at the bias reference plus V_D .

4.2 Clamping Circuits

A clamper (DC restorer) uses a diode and capacitor to shift the DC level while preserving the input's peak-to-peak. During one half-cycle the diode conducts and charges the capacitor to approximately the input peak (minus V_D); during the opposite half-cycle the diode is reverse-biased and the capacitor provides the DC shift. To maintain the waveform shape, the discharge through R over a period should be small: $RC \gg T = 1/f$.

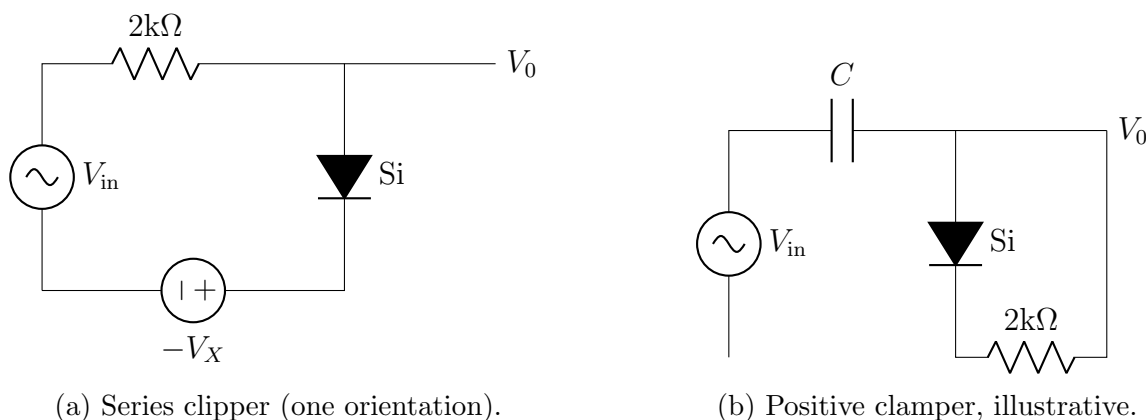


Figure 1: Representative clipper and clamper topologies (not to scale).

5 Equipment and Components

Multimeter, oscilloscope, function generator, DC supply; resistors: $2\text{ k}\Omega$ (1), $51\text{ k}\Omega$ (1); capacitor: $10\text{ }\mu\text{F}$ (1); diodes: 1N457 (2).

Note: Use available lab stock if the exact part number is unavailable; record substitutions.

6 Experimental Procedures

6.1 Clipping Circuits

1a) Biased clipper (-3 V reference). Build the circuit with $V_X = 3\text{ V}$ DC bias (polarity as assigned). Set input: square wave, $V_{pp} = 8\text{ V}$, $f = 1\text{ kHz}$. Measure and record V_0 for positive/negative input.

1b) Reverse diode, then add parallel clipper. Reverse diode orientation in 1a and repeat measurements. Then add a second diode in parallel as indicated by the lab handout; measure V_0 again for both polarities.

1c) Series clipper (with DC source). Build the series clipper variant. Record V_0 for positive/negative input. Reverse diode and repeat. Finally, determine the *minimum* DC magnitude required to guarantee conduction in reverse orientation (state assumptions).

6.2 Clamping Circuit

2a) DC restorer with bias. Build the clamper with -1.5 V bias. Input: square, 8 V_{pp} , 1 kHz . Record V_0 for both polarities.

2b) Frequency sweep. Repeat 2a at 50 kHz and 100 kHz . Capture input/output and comment on waveform shape vs frequency; relate to $\tau = RC$.

7 Data Recording Sheets

7.1 1a) Biased clipper (-3 V)

Table 1: Measured output levels, 1a

Condition	V_0 (meas.)
$V_{in} > 0$	_____
$V_{in} < 0$	_____

7.2 1b) Reverse diode

Table 2: Measured output levels, 1b (reversed)

Condition	V_0 (meas.)
$V_{in} > 0$	_____
$V_{in} < 0$	_____

7.3 1b) Parallel clippers

Table 3: Measured output levels, 1b (parallel)

Condition	V_0 (meas.)
$V_{in} > 0$	_____
$V_{in} < 0$	_____

7.4 1c) Series clipper

Table 4: Measured output levels, 1c

Condition	V_0 (meas.)
$V_{in} > 0$	_____
$V_{in} < 0$	_____

Minimum DC to guarantee conduction (reverse-orient): _____ Assumptions: _____

7.5 2a) Clamper (-1.5 V)

Table 5: Measured output levels, 2a

Condition	V_0 (meas.)
$V_{\text{in}} > 0$	_____
$V_{\text{in}} < 0$	_____

7.6 2b) Frequency sweep

Table 6: Observed output shape vs frequency (qualitative)

Frequency	Observation (shape/offset/droop)	Notes (RC rationale)
1 kHz	_____	_____
50 kHz	_____	_____
100 kHz	_____	_____

8 Analysis

8.1 Predicted Clip Levels

Assume silicon $V_D \approx 0.7\text{ V}$ unless measured otherwise. For each topology and polarity, compute the expected threshold V_0 when the diode conducts:

$$V_{\text{clip}} \approx \pm(V_X + n V_D)$$

where V_X is the DC bias magnitude (if present) and n is the number of series forward drops along the conduction path. Compare these predictions to measured V_0 .

8.2 Clamper Time-Constant Check

With $R = 2\text{ k}\Omega$ and $C = 10\text{ }\mu\text{F}$, $\tau = RC = 20\text{ ms}$. For $f = 1\text{ kHz}$, $T = 1\text{ ms}$ so $\tau/T \approx 20$. For 50 kHz and 100 kHz , $T = 20\text{ }\mu\text{s}$ and $10\text{ }\mu\text{s}$ respectively, yielding $\tau/T \gg 1$. Note any deviations due to source resistance, diode leakage, and ESR.

8.3 Scope Captures

Insert annotated screenshots for each subpart showing input and output with measured levels.

9 Optional: SPICE Netlists

9.1 Parallel/Series Biased Clipper (example)

```
* Biased series clipper (example values)
Vin in 0 PULSE(-4 4 0 1u 1u 0.5m 1m)
R1 in out 2k
Vx ref 0 DC 3
D1 out ref DSi
.model DSi D(Is=1e-14 N=1.9 Rs=2)
.tran 0 10m 0 1u
.probe V(in) V(out)
.end
```

9.2 Clamper

```
* Clamper (positive restorer example)
Vin in 0 PULSE(-4 4 0 1u 1u 0.5m 1m)
C1 in out 10u
D1 out 0 DSi
Rload out 0 2k
.model DSi D(Is=1e-14 N=1.9 Rs=2)
.tran 0 10m 0 1u
.probe V(in) V(out)
.end
```

10 Results

Summarize measured clip levels and DC shifts in tables, include uncertainties/estimation method (scope resolution, vertical scale), and attach representative captures.

11 Discussion

Comment on: (i) agreement between predicted and measured clip thresholds; (ii) effect of diode orientation and added parallel path; (iii) clamper performance vs frequency relative to RC ; (iv) non-idealities observed.

12 Conclusion

Provide a concise statement tying measurements to the theory of diode clipping and clamping. Note any design guidance (e.g., choosing V_X , accounting for V_D , selecting RC for a target frequency band).

Appendix

Attach any additional hand calculations, datasheet snippets for the diode used, and exported raw data.